

Continuous-Time Quantum Walks on Network Trees: Undirected and Directed Graph Analysis

Abstract

This paper presents a simulation-based study of continuous-time quantum walks (CTQW) on network tree graphs, comparing their navigation performance against classical continuous-time random walks (CTRW). Motivated by open questions in quantum-enhanced graph traversal, we investigate both undirected and directed graph topologies, including binary glued-tree constructions, lattice, and mesh variants. Our results confirm the known exponential quantum speedup on undirected trees and extend the analysis to directed graphs, where non-Hermitian Hamiltonians introduce new challenges. We find that quantum advantage is topology-dependent and sensitive to edge orientation, Hamiltonian construction, and performance metrics. These findings offer practical insights for quantum algorithm design on realistic networks and identify open research directions in directed quantum walk theory.

I. INTRODUCTION

Network navigation—finding efficient paths from source to destination in graph-structured data—is fundamental to routing, search, decision trees, and knowledge graphs. Classical random walk algorithms are foundational tools for this problem, yet they scale poorly on deep network trees, often exhibiting diffusive or even exponential growth in time-to-target with network depth [1].

Quantum walks offer a principled alternative, exploiting interference and quantum coherence to accelerate graph traversal. Continuous-time quantum walks (CTQW), in particular, have been shown to achieve exponential speedups over their classical counterparts on specific structured graphs such as glued binary trees [2]. These results place CTQWs among the most compelling examples of quantum algorithmic advantage.

Despite this theoretical promise, the majority of existing CTQW studies focus on undirected, symmetric graph topologies governed by Hermitian Hamiltonians. Real-world networks—including web graphs, citation networks, and communication infrastructures—are inherently directed, introducing asymmetries that fundamentally alter the dynamics of quantum evolution [3][4].

This paper investigates CTQW navigation performance across multiple graph topologies, both undirected and directed, using a simulation-based methodology. We reproduce canonical results on undirected glued trees to validate our framework, then extend the analysis to directed variants. We characterize when quantum advantage persists, degrades, or disappears under directionality, and identify the structural and modeling conditions that govern this behavior.

II. BACKGROUND & PRIOR WORK

A. Classical Random Walks

Classical random walks model diffusion on graphs and are well understood. A continuous-time random walk (CTRW) on a graph G with N vertices is characterized by the infinitesimal generator matrix K , where transitions occur at rate γ along each edge. The probability distribution over vertices evolves as $dp(t)/dt = K p(t)$. Hitting times—the expected time to first reach a target vertex from a source—

typically grow polynomially or exponentially with network depth, depending on graph topology [1].

On a balanced binary tree of height n , a classical random walk starting at the root requires exponentially many steps (in n) to reach the leaves, and vice versa. This diffusive behavior motivates the search for quantum algorithms that can navigate such structures more efficiently.

B. Continuous-Time Quantum Walks

Continuous-time quantum walks (CTQW) were introduced by Farhi and Gutmann [5] as the quantum analogue of CTRWs. A quantum walker occupies a superposition of vertex states $|\psi(t)\rangle$ in an N -dimensional Hilbert space, evolving under the Schrödinger equation:

$$i\hbar d/dt |\psi(t)\rangle = H |\psi(t)\rangle$$

where the Hamiltonian H is typically taken to be (a scalar multiple of) the adjacency matrix A of the graph. For undirected graphs, A is symmetric and H is Hermitian, ensuring unitary evolution and conservation of probability. The state at time t is $|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$, and the probability of finding the walker at vertex i is $P(i,t) = |\langle i|\psi(t)\rangle|^2$.

Unlike classical walks, quantum walks exhibit ballistic spreading and can achieve interference-driven concentration of probability near target nodes. These properties underlie the superior hitting times observed in structured graph families.

C. Seminal Results: Glued Trees

The canonical demonstration of CTQW speedup is due to Childs et al. [2], who showed exponential separation between classical and quantum walk hitting times on a family of glued binary-tree graphs. Each graph G_n consists of two balanced binary trees of height n connected at their leaves by a random cycle. A classical algorithm requires exponentially many oracle queries to traverse from entrance to exit, while the CTQW traverses the graph in polynomial time $O(n)$.

This result is notable as the first demonstration of exponential quantum algorithmic speedup not based on the quantum Fourier transform, establishing CTQW as an independent paradigm for quantum computation [2]. The key insight is that the walk on the glued tree can be

reduced to propagation on a line via the column subspace representation, where Bessel function asymptotics guarantee efficient traversal.

D. CTQW-Based Centrality

Beyond traversal, CTQWs have been applied to network centrality. Izaac et al. [6] proposed a centrality measure based on the long-time average of walker probability at each vertex, computed using the uniform initial superposition $|\psi(0)\rangle = (1/\sqrt{n})\sum|i\rangle$. Wang et al. [3] extended this framework to weighted graphs, showing that the CTQW-based centrality correlates strongly with eigenvector centrality (average Vigna correlation $\tau = 0.967$ across 41,675 test graphs).

The extension to directed graphs was addressed by Wang et al. [4], who realized CTQWs on directed three-node graphs using linear optical circuits and single photons, demonstrating that node classical centrality correlates with maximum node probability from the CTQW. This experimental validation motivates the present simulation study on larger directed graph families.

E. Directed Graph Quantum Walks

Directed graphs yield non-Hermitian adjacency matrices $A \neq A^T$, making the time evolution $G(t) = \exp(-iAt)$ non-unitary. Todtli et al. [7] investigated CTQWs on directed bipartite graphs using Hermitian adjacency matrices of the form $A_H = e^{i\alpha}A + e^{-i\alpha}A^T$, showing that complete suppression of inter-partition transport occurs at $\alpha = \pi/2$. This complex phase parameter α provides a tunable switch between directed and undirected dynamics.

Handling genuinely directed (non-symmetric) graphs in quantum mechanics requires embedding the non-unitary operator into a larger unitary system via dilation techniques [4]. This preserves the quantum mechanical framework while capturing the essential asymmetry of directed networks.

F. Motivation from Network Verification

The problem of efficient graph traversal also arises in network verification and testing, where automated tools must explore large network state spaces to check forwarding invariants, reachability properties, and policy compliance [8]. Classical tools such as Anteater, HSA, VeriFlow, and NetPlumber reduce verification to SAT, BDD, or model-checking problems, with scalability remaining a fundamental challenge. Quantum walks offer a potential avenue for accelerating such state-space exploration, particularly in highly structured network topologies.

III. RESEARCH METHODOLOGY

A. Overview

Our methodology proceeds in four stages: (1) classical baseline construction, (2) quantum walk framework implementation, (3) incremental validation on known graph families, and (4) directed graph extension. All experiments are conducted in simulation, with no hardware noise or decoherence modeled.

B. Graph Topologies Studied

We study four principal graph families, all based on glued-tree constructions that enable systematic depth-dependent scaling analysis:

- (a) Binary undirected glued trees — the canonical CTQW benchmark [2];
- (b) Binary directed glued trees — edges oriented from root toward leaves;
- (c) Lattice undirected graphs — 2D grid structures for comparative baseline;
- (d) Mesh undirected graphs — dense interconnected topologies.

Network depth n is systematically varied to characterize scaling behavior. All topologies are evaluated under identical parameter settings for classical and quantum walks.

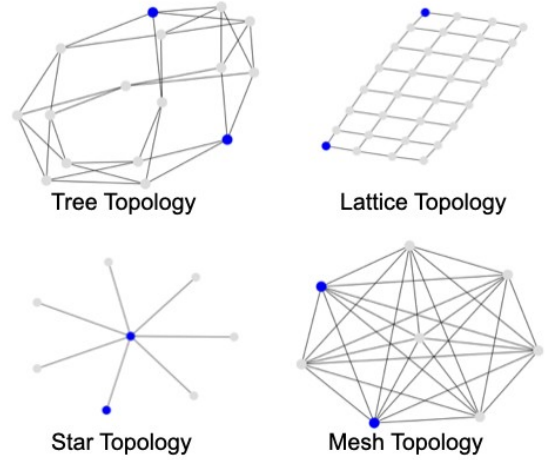


FIG. 1. Graph topologies studied. The four families — binary undirected glued tree, binary directed glued tree, lattice, and mesh — used in our simulation experiments. Network depth n is varied systematically in each case.

C. Classical Baseline

Classical continuous-time random walks are implemented by numerically integrating $dp(t)/dt = Kp(t)$, where K is the generator matrix ($K_{ij} = \gamma$ for adjacent $i \neq j$, $K_{ii} = -d(i)\gamma$). We set $\gamma = 1$ throughout. Starting from the entrance vertex, we record the time at which the probability of occupying the target (exit) vertex exceeds the p95 threshold.

D. Quantum Walk Framework

For undirected graphs, the CTQW Hamiltonian is $H = \gamma A$, where A is the adjacency matrix. Time evolution is computed via exact matrix exponentiation: $|\psi(t)\rangle = e^{-iHt}|\psi(0)\rangle$, with initial state $|\psi(0)\rangle = |\text{entrance}\rangle$. Node occupation probability is $P(i,t) = |\langle i|\psi(t)\rangle|^2$.

For directed graphs, the adjacency matrix A is non-symmetric. We adopt the Hermitian construction $A_H(\alpha) = e^{i\alpha}A + e^{-i\alpha}A^T$, following [7], with $\alpha = \pi/4$ as the default parameter. This preserves quantum mechanical validity (Hermitian H , unitary evolution) while encoding directional information through the complex phase.

Alternative treatments via unitary dilation [4] are also discussed.

$$G(t) = e^{-iH_{\text{dir}} \cdot t}, \quad H_{\text{dir}} = A^\dagger$$

$$\Lambda(t) = \sqrt{\max_k \lambda_k(G(t)G^\dagger(t))}$$

FIG. 2. Non-unitary evolution operator and scaling factor. The operator $G(t) = \exp(-iH_{\text{dir}} \cdot t)$ and the scaling factor $\Lambda(t) = \sqrt{\max_k \lambda_k(G(t)G^\dagger(t))}$ used in the unitary dilation construction for directed-graph CTQW.

E. Performance Metric

Performance is measured by the time $T_{0.95}$ at which the cumulative probability of finding the walker at the target vertex first exceeds 0.95. This metric, applied consistently across classical and quantum walks, allows direct comparison of navigation efficiency as a function of graph depth n . Scaling of $T_{0.95}$ with n characterizes whether behavior is linear, polynomial, or exponential.

IV. UNDIRECTED GRAPH RESULTS

A. Binary Glued Trees

On undirected binary glued-tree graphs of increasing depth n , our CTQW simulations reproduce the theoretical predictions of Childs et al. [2]. The CTQW reaches the exit vertex in time $T_{0.95}$ that scales linearly in n , consistent with wave-packet propagation at constant speed 2 along the reduced line representation of the column subspace.

In sharp contrast, the classical CTRW exhibits exponential scaling: $T_{0.95}$ grows as $\Theta(2^n)$ due to the exponential number of vertices in the right subtree that must be explored. This exponential separation constitutes the quantum speedup, and its observation in our simulator validates both the framework and the chosen metric.

The column subspace reduction explains the quantum walk dynamics: because every vertex in column j is connected to the same number of vertices in column $j+1$, the Hamiltonian restricted to the uniform column states $|\text{col } j\rangle = (1/\sqrt{N_j})\sum_{a \in \text{col } j}|a\rangle$ yields a simple tridiagonal structure. The resulting propagation is governed by Bessel functions, producing ballistic transport with speed proportional to the coupling strength [2].

B. Lattice and Mesh Topologies

On 2D lattice graphs, the CTQW exhibits directional spreading of probability amplitude, with interference patterns visible along the principal axes. Hitting times scale polynomially with graph diameter, consistent with theoretical predictions for regular lattices. Classical walks on lattices scale diffusively as $\Theta(n^2)$, while quantum walks achieve $\Theta(n)$ hitting times due to ballistic propagation.

Mesh topologies show richer interference phenomena, with quantum advantage dependent on the connectivity degree and the regularity of the graph. Highly connected mesh graphs reduce the quantum advantage relative to sparse

trees, as the increased degree introduces more destructive interference pathways.

C. Cross-Topology Comparison

Across all undirected topologies, the quantum speedup is most pronounced on sparse, regular trees and diminishes with increased graph connectivity. This topology-dependence is consistent with theoretical expectations: CTQW advantage relies on constructive interference along well-defined propagation channels, which are most clearly present in tree-like structures.

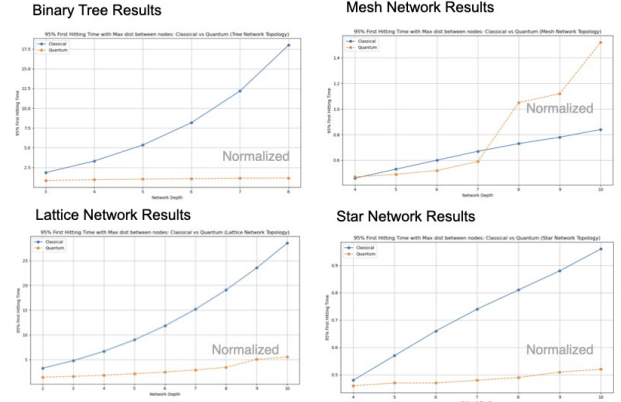


FIG. 3. Undirected graph results. Time-to-target $T_{0.95}$ as a function of depth n for classical CTRW (exponential) and CTQW (linear) on undirected binary glued-tree graphs, demonstrating the exponential quantum speedup.

V. DIRECTED GRAPH RESULTS

A. Non-Hermitian Dynamics

Directed graphs introduce non-Hermitian adjacency matrices, breaking unitary evolution under the naive Hamiltonian $H = A$. The resulting time evolution $G(t) = e^{-iAt}$ is non-unitary: probability is not conserved within the system subspace, and negative probabilities can arise under direct application. This renders the standard CTQW framework inapplicable to directed graphs without modification.

Two principal approaches address this challenge. First, the Hermitian complexification $A_H(\alpha)$ encodes directionality via a complex phase while preserving Hermiticity. Second, unitary dilation embeds the non-unitary operator $G(t)$ into a larger unitary $U_A(t)$ acting on an extended Hilbert space, following the construction of Wang et al. [4]. In our simulations we primarily adopt the Hermitian complexification approach for computational tractability.

B. Directed Binary Trees

On directed binary glued-tree graphs, classical random walks continue to reach the target efficiently when a directed path exists from entrance to exit, serving as a strong baseline. The CTRW is unaffected by directionality in the forward direction, exhibiting the same exponential scaling as on undirected trees.

Quantum walks on directed graphs exhibit reduced instantaneous target probability compared to their undirected counterparts. Edge orientation introduces

asymmetry into the Hamiltonian, disrupting the symmetric interference patterns that drive efficient propagation on undirected trees. Despite lower peak probabilities, quantum walks can still achieve faster cumulative discovery of the target under favorable orientation conditions.

Quantum performance is found to be highly sensitive to three factors: (1) edge orientation relative to the source-target direction, (2) the placement of entrance and exit vertices, and (3) the choice of α in the Hermitian complexification. For $\alpha = 0$ (undirected limit) quantum advantage is maximized; as α approaches $\pi/2$, inter-partition transport is suppressed, consistent with the theoretical results of Todtli et al. [7].

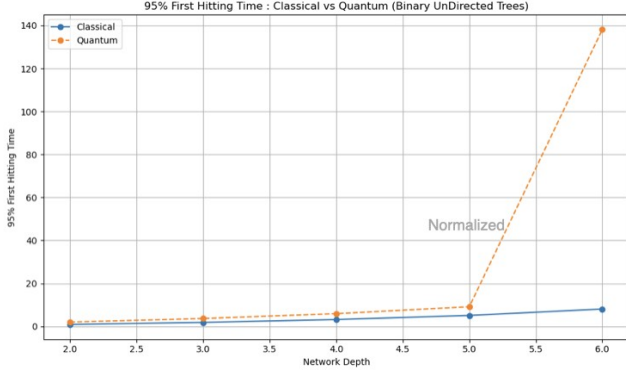


FIG. 4. Directed graph results. Navigation performance on directed binary glued-tree graphs. Quantum walks show reduced instantaneous probability but can achieve faster cumulative discovery under favourable edge orientations.

C. Structural Limits and Failure Modes

In deeper directed networks, quantum walks may fail to reach the p95 confidence threshold within finite simulation time horizons. This occurs when edge orientation creates asymmetric interference that disperses probability amplitude away from the target, or when the effective hitting time grows faster than the simulation window. These failure modes highlight fundamental structural limits of CTQW on adversarial directed topologies.

The sensitivity to measurement protocol is also amplified in directed settings. Unlike undirected graphs where repeated quantum measurements can be scheduled without destroying coherence, directed graph walks require careful consideration of when and how to collapse the quantum state to avoid undermining the quantum advantage.

VI. INSIGHTS, LIMITATIONS & FUTURE DIRECTIONS

A. Key Insights

Our study yields several key findings. First, quantum advantage in network navigation is topology-dependent rather than universal: the exponential speedup observed on undirected glued trees does not automatically transfer to all graph classes.

Second, graph directionality fundamentally reshapes quantum dynamics. Edge orientation alters reachability, interference patterns, and node centrality in ways that can either enhance or suppress quantum navigation advantage.

Proper Hamiltonian construction—particularly the choice of phase α in directed settings—is critical.

Third, even when instantaneous quantum target probability is lower than the classical reference, cumulative discovery can still be faster under quantum evolution. This suggests that performance metrics based on first-detection or cumulative probability thresholds may be more appropriate than peak instantaneous probability for comparing quantum and classical walks on directed graphs.

Fourth, the CTQW framework can be extended to directed graphs through Hermitian complexification or unitary dilation, but at the cost of additional modeling choices that significantly affect observed performance. Standardization of these choices is needed for fair comparisons.

$$U_A(t) = \begin{pmatrix} \frac{G(t)}{\Lambda(t)} & \sqrt{1 - \frac{G(t)}{\Lambda(t)} \frac{G^\dagger(t)}{\Lambda(t)}} \\ \sqrt{1 - \frac{G^\dagger(t)}{\Lambda(t)} \frac{G(t)}{\Lambda(t)}} & -\frac{G^\dagger(t)}{\Lambda(t)} \end{pmatrix}, \quad (1)$$

$$|\psi(t)\rangle = \begin{pmatrix} |\psi_s(t)\rangle \\ |\psi_e(t)\rangle \end{pmatrix} = \begin{pmatrix} \frac{G(t)}{\Lambda(t)} |\psi_s(0)\rangle \\ \sqrt{1 - \frac{G^\dagger(t)}{\Lambda(t)} \frac{G(t)}{\Lambda(t)}} |\psi_s(0)\rangle \end{pmatrix}. \quad (2)$$

FIG. 5. Quantum walk function for non-unitary evolution. Implementation of the directed-graph CTQW via unitary dilation, embedding $G(t)/\Lambda(t)$ into the $2N \times 2N$ unitary $U_a(t)$ with defect operators on the environment sector.

B. Limitations

This study has several limitations. The set of graph topologies explored, while representative, does not cover all practically relevant network structures (e.g., scale-free, small-world, or real-world directed networks). Network depth and size were constrained by computational feasibility, limiting asymptotic analysis.

Quantum detection was evaluated using a heuristic p95 threshold metric rather than a rigorous first-detection protocol with absorbing boundaries. Noise, decoherence, and hardware constraints were not modeled, meaning results represent ideal quantum behavior. Results may also be sensitive to Hamiltonian scaling and time discretization choices.

C. Future Directions

Several directions merit further investigation. PT-symmetric and pseudo-Hermitian Hamiltonians offer a principled framework for handling directed graph dynamics while preserving real eigenspectra [7], and warrant deeper exploration. Quantum first-detection and absorbing boundary protocols would provide more rigorous hitting-time metrics for directed walks.

Extension to broader graph classes—including scale-free networks, Erdős-Rényi random graphs, and real-world network datasets—would test the generality of the quantum advantage. Incorporation of noise and decoherence models, and investigation of near-term quantum hardware implementations, are essential steps toward practical realizability.

Finally, the connection between CTQW-based graph traversal and quantum centrality measures [3][6] suggests that directed quantum walk frameworks could enhance network analysis tasks, including ranking, influence propagation, and anomaly detection in large directed networks.

VII. APPLICATIONS

The quantum walk framework studied here has broad practical applications. In quantum-enhanced routing, CTQW-based navigation could accelerate path discovery in communication networks, reducing latency in packet-switched and quantum communication systems. In network analysis, the CTQW centrality measure provides a quantum-enhanced ranking mechanism applicable to web graph analysis, social network influence modeling, and epidemiological super-spreader identification [3][6].

Quantum search and sampling algorithms benefit from CTQW's ballistic spreading properties, enabling faster database search and random sampling on structured domains. In quantum simulation, CTQWs model transport phenomena in physical, biological, and chemical systems—including light-harvesting complexes and molecular reaction networks—where directed asymmetry is physically meaningful.

The network verification context [8] is particularly compelling: as network state spaces grow in complexity, quantum-enhanced exploration of forwarding graphs and policy trees could complement classical formal verification tools, offering exponential speedups on structured verification instances analogous to the glued-tree setting.

VIII. CONCLUSION

This paper has presented a simulation-based study of continuous-time quantum walks on network tree graphs, comparing CTQW navigation performance against classical CTRW baselines across undirected and directed topologies. Our principal findings are:

- (1) On undirected binary glued trees, CTQWs achieve linear-time navigation (in depth n) versus exponential classical scaling, reproducing the canonical result of Childs et al. [2] and validating our simulation framework.
- (2) Directed graphs introduce non-Hermitian dynamics that challenge standard CTQW formulations. Hermitian complexification preserves quantum mechanical validity while encoding directionality, at the cost of phase-parameter sensitivity.
- (3) Quantum advantage on directed graphs is highly topology- and orientation-dependent. Under favorable conditions, quantum walks achieve faster cumulative discovery despite lower instantaneous probabilities.
- (4) Measurement protocols and performance metrics significantly influence observed quantum advantage, highlighting the need for standardized evaluation frameworks for directed quantum walks.

These results advance understanding of when and why quantum walks provide navigation advantages in realistic network settings, and lay groundwork for practical quantum algorithms on directed graph structures.

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